

OpenStax Calculus Volume 2

Section 2.9 Exercises: Calculus of the Hyperbolic Functions

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1. [T] Find expressions for $\cosh x + \sinh x$ and $\cosh x - \sinh x$. Use a calculator to graph these functions and ensure your expression is correct.
2. From the definitions of $\cosh(x)$ and $\sinh(x)$, find their antiderivatives.
3. Show that $\cosh(x)$ and $\sinh(x)$ satisfy $y'' = y$.
4. Use the quotient rule to verify that $\tanh(x)' = \operatorname{sech}^2(x)$.
5. Derive $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$ from the definition.
6. Take the derivative of the previous expression to find an expression for $\sinh(2x)$.
7. Prove $\sinh(x+y) = \sinh(x) \cosh(y) + \cosh(x) \sinh(y)$ by changing the expression to exponentials.
8. Take the derivative of the previous expression to find an expression for $\cosh(x+y)$.

For the following exercises, find the derivatives of the given functions and graph along with the function to ensure your answer is correct.

9. [T] $\cosh(3x + 1)$
10. [T] $\sinh(x^2)$
11. [T] $\frac{1}{\cosh(x)}$
12. [T] $\sinh(\ln(x))$
13. [T] $\cosh^2(x) + \sinh^2(x)$
14. [T] $\cosh^2(x) - \sinh^2(x)$
15. [T] $\tanh(\sqrt{x^2 + 1})$
16. [T] $\frac{1+\tanh(x)}{1-\tanh(x)}$
17. [T] $\sinh^6(x)$
18. [T] $\ln(\operatorname{sech}(x) + \tanh(x))$

For the following exercises, find the antiderivatives for the given functions.

19. $\cosh(2x + 1)$
20. $\tanh(3x + 2)$
21. $x \cosh(x^2)$
22. $3x^3 \tanh(x^4)$
23. $\cosh^2(x) \sinh(x)$
24. $\tanh^2(x) \operatorname{sech}^2(x)$
25. $\frac{\sinh(x)}{1+\cosh(x)}$
26. $\coth(x)$
27. $\cosh(x) + \sinh(x)$
28. $(\cosh(x) + \sinh(x))^n$

For the following exercises, find the derivatives for the functions.

29. $\tanh^{-1}(4x)$
30. $\sinh^{-1}(x^2)$
31. $\sinh^{-1}(\cosh(x))$
32. $\cosh^{-1}(x^3)$
33. $\tanh^{-1}(\cos(x))$
34. $e^{\sinh^{-1}(x)}$
35. $\ln(\tanh^{-1}(x))$

For the following exercises, find the antiderivatives for the functions.

36. $\int \frac{dx}{4-x^2}$
37. $\int \frac{dx}{a^2-x^2}$
38. $\int \frac{dx}{\sqrt{x^2+1}}$
39. $\int \frac{x dx}{\sqrt{x^2+1}}$
40. $\int -\frac{dx}{x\sqrt{1-x^2}}$
41. $\int \frac{e^x}{\sqrt{e^{2x}-1}} dx$
42. $\int -\frac{2x}{x^4-1} dx$

For the following exercises, use the fact that a falling body with friction equal to velocity squared obeys the equation $dv/dt = g - v^2$.

43. Show that $v(t) = \sqrt{g} \tanh((\sqrt{g})t)$ satisfies this equation.
44. Derive the previous expression for $v(t)$ by integrating $\frac{dv}{g-v^2} = dt$.

45. [T] Estimate how far a body has fallen in 12 seconds by finding the area underneath the curve of $v(t)$.

For the following exercises, use this scenario: A cable hanging under its own weight has a slope $S = dy/dx$ that satisfies $dS/dx = c\sqrt{1 + S^2}$. The constant c is the ratio of cable density to tension.

46. Show that $S = \sinh(cx)$ satisfies this equation.
47. Integrate $dy/dx = \sinh(cx)$ to find the cable height $y(x)$ if $y(0) = 1/c$.
48. Sketch the cable and determine how far down it sags at $x = 0$.

For the following exercises, solve each problem.

49. [T] A chain hangs from two posts 2 m apart to form a catenary described by the equation $y = 2 \cosh(x/2) - 1$. Find the slope of the catenary at the left fence post.
50. [T] A chain hangs from two posts four meters apart to form a catenary described by the equation $y = 4 \cosh(x/4) - 3$. Find the total length of the catenary (arc length).
51. [T] A high-voltage power line is a catenary described by $y = 10 \cosh(x/10)$. Find the ratio of the area under the catenary to its arc length. What do you notice?
52. A telephone line is a catenary described by $y = a \cosh(x/a)$. Find the ratio of the area under the catenary to its arc length. Does this confirm your answer for the previous question?
53. Prove the formula for the derivative of $y = \sinh^{-1}(x)$ by differentiating $x = \sinh(y)$. (Hint: Use hyperbolic trigonometric identities.)
54. Prove the formula for the derivative of $y = \cosh^{-1}(x)$ by differentiating $x = \cosh(y)$. (Hint: Use hyperbolic trigonometric identities.)
55. Prove the formula for the derivative of $y = \operatorname{sech}^{-1}(x)$ by differentiating $x = \operatorname{sech}(y)$. (Hint: Use hyperbolic trigonometric identities.)
56. Prove that $(\cosh(x) + \sinh(x))^n = \cosh(nx) + \sinh(nx)$.
57. Prove the expression for $\sinh^{-1}(x)$. Multiply $x = \sinh(y) = (1/2)(e^y - e^{-y})$ by $2e^y$ and solve for y . Does your expression match the textbook?
58. Prove the expression for $\cosh^{-1}(x)$. Multiply $x = \cosh(y) = (1/2)(e^y + e^{-y})$ by $2e^y$ and solve for y . Does your expression match the textbook?